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UNIVERSITY OF SCIENCE  
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# COMP 5212

# Machine Learning

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Lecture 2: Hypothesis Sets and PAC Learning

# Learning Goals

By the end of this lecture, you should be able to:

- understand the concept of a **hypothesis set**.
- distinguish **empirical** from **generalization** error;
- state the basic **PAC-learning guarantee**;
- derive the finite-class sample bound using a union bound;

# Binary Classification

## Case 1: Near-deterministic label — Is there a dog?

- Given input  $x$ , the label  $y \in \{0, 1\}$  is almost completely determined.
- The same  $x$  essentially always maps to the same  $y$ .
- Noise structure: low uncertainty;
- Model implication: a deterministic hypothesis  $h : \mathcal{X} \rightarrow \{0, 1\}$  is appropriate.
- We can learn a sharp decision boundary and minimise classification error.



# Binary Classification

## Case 2: Stochastic label — Will it rain today?

- The same cloudy sky  $x$  can occur on both rainy and non-rainy days.
- Input  $x$  misses important predictors: humidity, pressure, wind, etc.
- Noise structure: high uncertainty;
- Model implication: use a probabilistic hypothesis  $h : \mathcal{X} \rightarrow [0, 1]$ , giving  $P(y = 1 | x)$ .
- Optimise proper scoring rules (log loss, Brier score); calibration matters.



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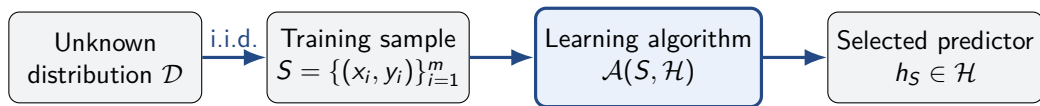
# Binary Classification

- This lecture introduces the foundational theory of **deterministic** binary classification.
- Why start with this case?
  - Cleaner mathematical structure  $\Rightarrow$  easier to derive sharp learning guarantees.
  - Serves as the essential building block for more complex stochastic and probabilistic settings.

# Hypothesis Sets

What predictors may the learning algorithm choose?

# From Data to a Predictor



Data select a predictor; the hypothesis set determines what can be selected.

# Formal Setup

- $\mathcal{X}$ : sample space.
- $\mathcal{Y} = \{0, 1\}$ : label space.
- $c : \mathcal{X} \rightarrow \mathcal{Y}$ : target concept.
- $\mathcal{C}$ : possible target concepts.
- $\mathcal{D}$ : unknown distribution on  $\mathcal{X}$ .

A labeled sample is

$$S = ((x_1, y_1), \dots, (x_m, y_m)), \quad y_i = c(x_i),$$

with

$$x_1, \dots, x_m \stackrel{\text{i.i.d.}}{\sim} \mathcal{D}.$$

The learner sees  $S$ , but not  $c$  or  $\mathcal{D}$ .

Training and test inputs are assumed to follow the same distribution  $\mathcal{D}$ .



# What Is a Hypothesis Set?

A **hypothesis set** (or hypothesis class) is a collection

$$\mathcal{H} \subseteq \{h : \mathcal{X} \rightarrow \mathcal{Y}\}$$

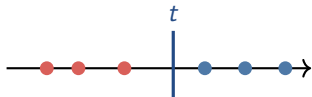
of candidate prediction rules available to the learner.

- The learning algorithm outputs  $h_S \in \mathcal{H}$ .
- $\mathcal{H}$  encodes an **inductive bias**: which explanations are considered plausible.
- Parameters index hypotheses; the entire parameterized family is the set  $\mathcal{H}$ .

Example:  $\mathcal{H} = \{x \mapsto \mathbf{1}\{w^T x + b \geq 0\} : w \in \mathbb{R}^d, b \in \mathbb{R}\}$ .

# Examples of Hypothesis Sets

Thresholds on a line



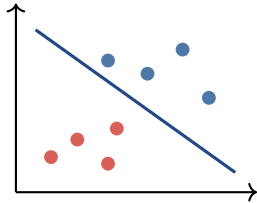
$$h_t(x) = \mathbf{1}\{x \geq t\}$$

Intervals



$$h_{a,b}(x) = \mathbf{1}\{a \leq x \leq b\}$$

Linear separators



$$h_{w,b}(x) = \mathbf{1}\{w^T x + b \geq 0\}$$

A hypothesis set may be finite or infinite; its capacity, not just its parameter count, controls generalization.

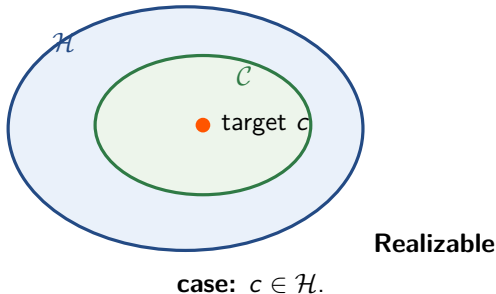
# Concept Class vs. Hypothesis Set

## Concept class $\mathcal{C}$

Describes possible data-generating target rules  $c$ .

## Hypothesis set $\mathcal{H}$

Describes predictors the algorithm is allowed to return.

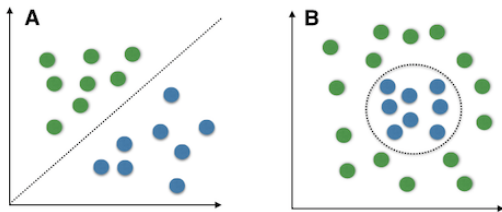


If  $c \notin \mathcal{H}$ , the learner can only find the best approximation available inside  $\mathcal{H}$ ; this is the **non-realizable** or **agnostic** setting.

# Inductive Bias

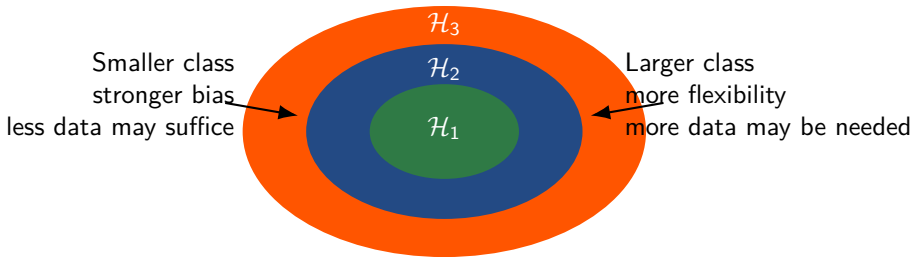
**Inductive bias:** How much prior knowledge or experience is used to constrain the set of plausible models?

**Example: Line v.s. Circle**



# Inductive Bias

**Inductive bias:** How much prior knowledge or experience is used to constrain the set of plausible models?



Too small  $\mathcal{H}$ : approximation error and underfitting.

Too rich  $\mathcal{H}$ : more data-fitting explanations and overfitting risk.

# ERM and Consistency

A learning algorithm maps a sample to a hypothesis:

$$\mathcal{A} : S \mapsto h_S \in \mathcal{H}.$$

## Empirical risk minimization (ERM)

$$h_S \in \arg \min_{h \in \mathcal{H}} \hat{R}_S(h).$$

Choose the smallest training error.

## Consistent learning

$$\hat{R}_S(h_S) = 0.$$

In a realizable problem, ERM can be consistent.

When does small training error imply small test error?

# Empirical Error vs. Generalization Error

## True error

$$R(h) = \Pr_{x \sim \mathcal{D}}[h(x) \neq c(x)].$$

Performance on future examples.

## Empirical error

$$\hat{R}_S(h) = \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{h(x_i) \neq c(x_i)\}.$$

Performance on the training sample.

For a fixed  $h$ ,  $\hat{R}_S(h)$  estimates  $R(h)$ . The subtlety is that the selected  $h_S$  depends on  $S$ .

# The Realizable Case

The basic PAC model begins with a favorable assumption:

$$c \in \mathcal{H}.$$

For example,  $\mathcal{C} = \mathcal{H}$ .

## Consequence

A zero-training-error hypothesis exists for every sample.

## Question

How large must  $m$  be so that a consistent hypothesis has error at most  $\epsilon$ ?

This leads to the Probably Approximately Correct framework.



# PAC Learning

A distribution-free, finite-sample notion of learnability

# PAC in Plain Language

## Probably

With probability at least  $1 - \delta$  over the random training sample.

## Approximately

The returned predictor may not be perfect, but its error is at most  $\epsilon$ .

## Correct

Correctness is measured on new examples drawn from the unknown  $\mathcal{D}$ .

PAC learning turns “learns well” into a statement involving  $m$ ,  $\epsilon$ , and  $\delta$ .

# PAC-Learnability: Formal Definition

A concept class  $\mathcal{C}$  is **PAC-learnable** w.r.t. a hypothesis set  $\mathcal{H}$ , if there exist a learner  $\mathcal{A}$  and a polynomial sample bound  $m_{\mathcal{C}}$  such that, for every  $c \in \mathcal{C}$ , distribution  $\mathcal{D}$ , and  $\varepsilon, \delta \in (0, 1)$ , whenever  $m \geq m_{\mathcal{C}}(\varepsilon, \delta)$ , it holds that  $\mathcal{A}(S) \in \mathcal{H}$  and

$$\Pr_{S \sim \mathcal{D}^m} [R(\mathcal{A}(S)) \leq \varepsilon] \geq 1 - \delta.$$

## Universal

The guarantee holds for every target and every input distribution.

## Probably

Probability is over the random training sample  $S$ .

## Approximately correct

The returned error is at most  $\varepsilon$ .

# What the Basic PAC Model Assumes

## Assumed

- examples are i.i.d.;
- training and test inputs share  $\mathcal{D}$ ;
- the basic model is realizable:  $c \in \mathcal{H}$ ;
- labels are binary in this presentation.

## Not assumed

- a known parametric form for  $\mathcal{D}$ ;
- Gaussian inputs or balanced classes;
- a particular learning algorithm;
- a specific geometry of  $\mathcal{X}$ .

PAC is **distribution-free**, but it does not cover arbitrary distribution shift.

# How to Read $\varepsilon$ and $\delta$

## Accuracy $\varepsilon$

We want

$$R(h_S) \leq \varepsilon.$$

Smaller  $\varepsilon$  means a more accurate predictor and usually requires more data.

## Failure probability $\delta$

We want the guarantee to hold with probability

$$1 - \delta.$$

Smaller  $\delta$  means higher confidence and usually requires more data.

Example:  $\varepsilon = 0.05$ ,  $\delta = 0.01$  means error at most 5% with at least 99% confidence.

# Sample Complexity

The **sample complexity**  $m_{\mathcal{H}}(\varepsilon, \delta)$  is the number of labeled examples sufficient for a PAC guarantee over  $\mathcal{H}$ .

It records how data requirements scale with:

- desired accuracy  $1/\varepsilon$ ;
- desired confidence  $\ln(1/\delta)$ ;
- the complexity of  $\mathcal{H}$ ;
- realizable versus agnostic assumptions.

Efficient PAC learning additionally requires polynomial running time.

# Finite $\mathcal{H}$ : Consistent Learning Bound

## Assume:

- $\mathcal{H}$  is finite;
- $c \in \mathcal{H}$ ;
- the learner returns  $h_S$  with  $\hat{R}_S(h_S) = 0$ .

With probability at least  $1 - \delta$ ,

$$R(h_S) \leq \frac{\ln |\mathcal{H}| + \ln(1/\delta)}{m}.$$

Thus, to ensure  $R(h_S) \leq \varepsilon$ , it suffices that

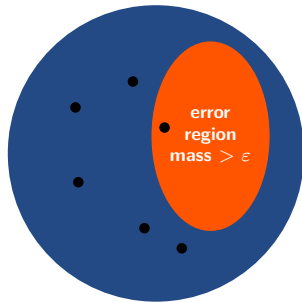
$$m \geq \frac{\ln |\mathcal{H}| + \ln(1/\delta)}{\varepsilon}.$$

# Proof Idea I: A Bad Hypothesis Is Unlikely to Look Perfect

Fix  $h$  with  $R(h) > \varepsilon$ . Each example falls in the error region with probability greater than  $\varepsilon$ .

$$\Pr[\hat{R}_S(h) = 0] \leq (1 - \varepsilon)^m \leq e^{-m\varepsilon}.$$

A bad hypothesis can look perfect only if all  $m$  examples miss its error region.





# Proof Idea II: Union Bound over All Bad Hypotheses

Define

$$\mathcal{H}_\varepsilon = \{h \in \mathcal{H} : R(h) > \varepsilon\}.$$

The learner fails only if some bad hypothesis is consistent:

$$\begin{aligned}\Pr\left[\exists h \in \mathcal{H}_\varepsilon : \hat{R}_S(h) = 0\right] &\leq \sum_{h \in \mathcal{H}_\varepsilon} \Pr\left[\hat{R}_S(h) = 0\right] \\ &\leq |\mathcal{H}| e^{-m\varepsilon}.\end{aligned}$$

Set  $|\mathcal{H}| e^{-m\varepsilon} \leq \delta$  and solve:

$$m \geq \frac{\ln |\mathcal{H}| + \ln(1/\delta)}{\varepsilon}.$$

# Finite-Class Bound

| Dependence          | Interpretation                                                     |
|---------------------|--------------------------------------------------------------------|
| $1/\varepsilon$     | Halving the target error roughly doubles the required sample size. |
| $\ln(1/\delta)$     | High confidence costs only logarithmically more data.              |
| $\ln  \mathcal{H} $ | The number of candidates enters through its logarithm.             |
| <b>consistency</b>  | The fast $1/m$ rate uses realizability and zero training error.    |

$\ln_2 |\mathcal{H}|$  is the number of bits needed to identify a hypothesis: a formal version of **Occam's razor**.

# Example: Discretized Thresholds

Suppose  $|\mathcal{H}| = 1001$  candidate thresholds and we want  $\varepsilon = 0.05$ ,  $\delta = 0.05$ .

$$m \geq \frac{\ln(1001) + \ln(20)}{0.05} \approx 198.1.$$

This is a worst-case guarantee, not a prediction of the exact data needed in a particular dataset.

# Example: Conjunctions of Boolean Literals

Let  $x \in \{0, 1\}^n$ . A conjunction can include each variable positively, negatively, or not at all:

$$h(x) = x_1 \wedge \neg x_4 \wedge x_7.$$

Thus there are at most

$$|\mathcal{H}| = 3^n$$

conjunctions, and consistent PAC learning needs

$$m \geq \frac{n \ln 3 + \ln(1/\delta)}{\epsilon}.$$

For  $n = 20$ ,  $\epsilon = 0.10$ , and  $\delta = 0.05$ , the bound gives  $m \geq 250$ .

# Beyond the Realizable Case

Noisy labels, model mismatch, and uniform convergence

# When No Hypothesis Is Perfectly Consistent

Real data may violate  $c \in \mathcal{H}$ :

- labels can be noisy or stochastic;
- the chosen model family may be misspecified;
- several labels may be plausible for the same input.

In the **agnostic setting**, compare the learner to the best hypothesis in  $\mathcal{H}$ :

$$h^* \in \arg \min_{h \in \mathcal{H}} R(h).$$

The goal is small **excess risk**

$$R(h_S) - R(h^*).$$

We now need training and true errors to be close *uniformly* over  $\mathcal{H}$ .

# Hoeffding's Inequality

For a fixed  $h$ , the 0–1 losses are independent Bernoulli random variables. Hoeffding's inequality:

$$\Pr\left[\left|R(h) - \widehat{R}_S(h)\right| > \gamma\right] \leq 2e^{-2m\gamma^2}.$$

## Caution

The output  $h_S$  is **not fixed**; it depends on the same sample  $S$  used to compute  $\widehat{R}_S(h_S)$ .

We need one event on which the bound holds simultaneously for every  $h \in \mathcal{H}$ .

# Concentration Inequalities

- Concentration inequalities is a class of probability bounds for sequential independent random variables.
- Why “Concentration”? As sample size grows, random variables cluster tightly around their expected value.
- **Core Usage:** Bounds the probability that a sample average deviates from its true mean. Essential for proving generalization bounds in PAC learning.



# Concentration Inequalities

- **Hoeffding's Inequality** (For bounded independent variables  $X_i \in [a_i, b_i]$ )

$$\mathbb{P} \left( \frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X] \geq \epsilon \right) \leq \exp \left( - \frac{2n^2 \epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2} \right)$$

- **Chernoff Bound** (For independent binary or multiplicative variables; sharper tails)

$$\mathbb{P} \left( \frac{1}{n} \sum_{i=1}^n X_i \geq (1 + \delta)\mu \right) \leq \left( \frac{e^\delta}{(1 + \delta)^{(1+\delta)}} \right)^{n\mu}$$

- **McDiarmid's Inequality** (For general functions with bounded differences  $c_i$ )

$$\mathbb{P} (f(X_1, \dots, X_n) - \mathbb{E}[f] \geq \epsilon) \leq \exp \left( - \frac{2\epsilon^2}{\sum_{i=1}^n c_i^2} \right)$$

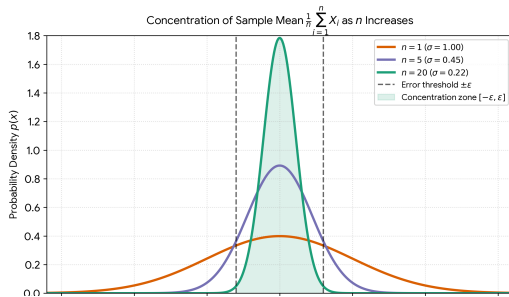
# Examples: Gaussian Noise

Assume  $X_1, X_2, \dots, X_n$  are i.i.d. Gaussian variables with mean  $\mu$  and variance  $\sigma^2$ , then  $\frac{1}{n} \sum_{i=1}^n X_i$  is a Gaussian variable with mean  $\mu$  and variance  $\sigma^2/n$ . Then

$$\mathbb{P} \left[ \left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| \geq \epsilon \right] \leq 2 \exp \left( -\frac{n\epsilon^2}{2\sigma^2} \right)$$

## Visualizing the Collapse of Variance:

- As  $n$  increases, the peak grows taller and narrower.
- The probability mass outside the  $\pm\epsilon$  boundaries drops to zero exponentially fast.
- This captures the physical essence of the “concentration of measure.”



# Uniform Convergence for Finite $\mathcal{H}$

Apply Hoeffding to every  $h \in \mathcal{H}$  with  $\gamma = \sqrt{\ln(2|\mathcal{H}|/\delta)/(2m)}$  and use a union bound.

With probability at least  $1 - \delta$ ,

$$\forall h \in \mathcal{H}, \quad \left| R(h) - \hat{R}_S(h) \right| \leq \sqrt{\frac{\ln(2|\mathcal{H}|/\delta)}{2m}}.$$

Uniform convergence lets us compare all hypotheses using the same training sample.

# ERM Guarantee

Let  $\hat{h}$  minimize empirical risk, let  $h^*$  minimize true risk over  $\mathcal{H}$ , and define

$$\gamma_m = \sqrt{\frac{\ln(2|\mathcal{H}|/\delta)}{2m}}.$$

On the uniform-convergence event,

$$R(\hat{h}) \leq \hat{R}_S(\hat{h}) + \gamma_m \leq \hat{R}_S(h^*) + \gamma_m \leq R(h^*) + 2\gamma_m.$$

$$R(\hat{h}) \leq \min_{h \in \mathcal{H}} R(h) + 2\sqrt{\frac{\ln(2|\mathcal{H}|/\delta)}{2m}}.$$

# Approximation and Estimation

Approximation error

Best achievable risk inside  $\mathcal{H}$

Reduce by enlarging or redesigning  $\mathcal{H}$ .

Estimation error

Price of selecting from finite data

Reduce with more data or lower-capacity  $\mathcal{H}$ .

Model selection balances the two: a richer hypothesis set improves fit but increases the amount of evidence needed for reliable selection.

# What About Infinite Hypothesis Sets?

Many useful classes are infinite:

- real-valued thresholds;
- intervals with real endpoints;
- linear separators in  $\mathbb{R}^d$ .

## Problem

The finite-class term  $\ln |\mathcal{H}|$  is no longer useful.

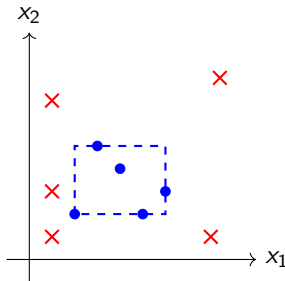
Measure how many distinct labelings  $\mathcal{H}$  can realize on finite samples. This leads to lecture 3: **growth functions**, **VC dimension**, and **Rademacher complexity**.

# Proper Learning

**Definition:** The learner must output a hypothesis  $h$  from the *same* class as the target concept class  $C$ , i.e.,  $H = C$ .

**Example:** Learning axis-aligned rectangles in  $\mathbb{R}^2$

- Target concept is an axis-aligned rectangle.
- Learner must output an axis-aligned rectangle that fits the training data.

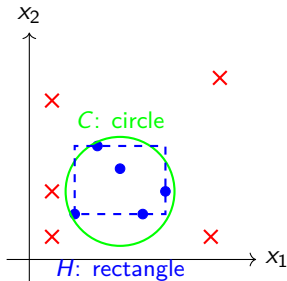


# Improper Learning

**Definition:** The learner may output a hypothesis  $h$  from a *different* (often larger) class  $H \neq C$ .

**Example:** Target concept is a circle, but learner outputs an axis-aligned rectangle

- $C = \text{circles in } \mathbb{R}^2$ .
- $H = \text{axis-aligned rectangles}$ .
- The rectangle separates the data equally well, even though  $H \neq C$ .





# Proper vs. Improper Learning

Trade-off between *computational tractability* and *generalization performance*.

## Proper Learning

- Higher interpretability
- Matches domain constraints
- Easier to validate
- Can be computationally harder (NP-hard)

## Improper Learning

- Greater computational efficiency
- Can bypass representation barriers
- Larger hypothesis space complexity
- Lower model interpretability

**Core Takeaway:** Improper learning expands the search space to find solutions faster, while proper learning keeps the model strictly interpretable.

# Proper vs. Improper Learning

## Intersections of Halfspaces

**The Target:** A bounded geometric polygon.

- **Proper Approach:** Forces exact linear boundary equations. Brutal and **NP-hard** to solve.
- **Improper Approach:** Outputs a **Decision Tree** approximating the shape. Efficient and fast.

**Core Takeaway:** Forcing a model to output the exact mathematical structure of the target concept often makes learning impossible in practice.

# Key Takeaways

- $\mathcal{H}$  is the learner's allowable predictor family and its inductive bias.
- PAC means  $R(h_S) \leq \varepsilon$  with probability at least  $1 - \delta$ .
- Finite, realizable classes need  $m = O((\ln |\mathcal{H}| + \ln(1/\delta))/\varepsilon)$  examples.
- In the agnostic case, ERM excess risk is  $O(\sqrt{(\ln |\mathcal{H}| + \ln(1/\delta))/m})$ .
- Infinite classes require capacity measures beyond cardinality, such as VC dimension or Rademacher complexity.

# Check Your Understanding

1. If  $|\mathcal{H}|$  doubles, how much does the finite realizable sample bound increase?
2. Why can a fixed- $h$  Hoeffding bound not be applied directly to the ERM output  $h_S$ ?
3. Give an example with  $c \notin \mathcal{H}$ . What is the correct comparison target?
4. Which is more expensive in the consistent bound: dividing  $\varepsilon$  by 10, or dividing  $\delta$  by 10?

# References and Further Reading



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Original instructional summary; definitions and finite-class bounds follow Chapter 2 of Mohri et al.